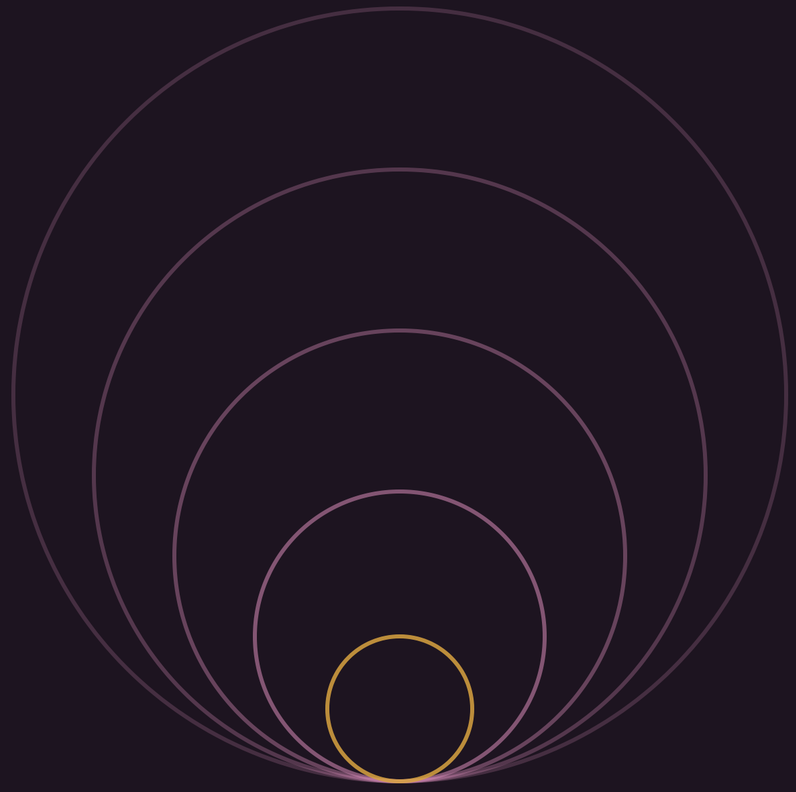
Kanda Code | कांदा

When should onion farmers sell their crop?



A price forecast and a linear programme decide, month by month, how much of a 1,000-tonne harvest to sell or store, tested on eight seasons of real prices.

Raghuram Devarayi · Parashmani Datta · Kathan Desai · Aaryan Choudhary · Yash Arabhavi

THE PROBLEM

Farmers sell at the year's lowest price

FARM GATE VS SHOP, MAY 2026

Rs 5–7/kg

paid to farmers while retail averaged Rs 25.36/kg (Outlook Business)

LASALGAON, 2022

Rs 7–10/kg

received by farmers against a cultivation cost of Rs 22–25/kg (Down To Earth)

SEASONAL INDEX, 2014–24

0.55× → 1.55×

April price vs November price, each as a multiple of that year's average

⚠ **Why not just wait? Stored onions rot.**

A ventilated kanda chawl loses 54.9% of its onions in 120 days; a cold store loses 5.0–7.1% (ICAR trials). Each month, the farmer must judge whether the price rise beats the rot.

WHY THIS PROBLEM

A yearly, costly, data-rich decision

SCALE

6.4 lakh t

It recurs every year, at scale

procured for the government onion buffer in 2023–24; target 5 lakh t for rabi 2024 (Krishi Jagran)

STORAGE LOSS

≈53%

Storage is where value is lost

of NAFED's 57,372 t buffer was lost in traditional storage in 2019 (ThePrint)

PERISHABILITY

4–6 months

Onion is the hardest case

it stores for months but rots fast; grain stores well and tomato not at all

INDUSTRY 4.0

54.9% vs 7.1%

A rupee value for smart storage

120-day loss in ventilated vs controlled cold storage (ICAR) sets the value of monitoring

RESEARCH QUESTIONS

Three questions, plus one extension

#	QUESTION	HYPOTHESIS	TEST
RQ1	Do heavier market arrivals push the price down?	**H1** · price falls as arrivals rise	Log-log OLS with month and year dummies; HAC errors
RQ2	Does storing in an ordinary chawl beat selling all at harvest?	**H2** · an optimised chawl plan earns more	LP plan vs harvest sale, scored at actual prices, 2017–24
RQ3	What is low-loss (cold / IoT) storage worth?	**H3** · it earns materially more per tonne than a chawl	Same LP, scenario C vs B; paired tests over 8 seasons
EXT	Does late-monsoon rain (NASA satellite data) predict the autumn price rise?	**H4** · more Sep–Oct rain, bigger rise	OLS over 29 seasons, 1996–2024; added beyond the proposal

OUR METHOD, ONE LAYER AT A TIME

Data · validate → Pattern · H1 → Forecast · Apr–Dec → Plan · LP → Value · backtest

DATA

Lasalgaon was missing. We rebuilt it.

THE GAP

0 rows

for Lasalgaon among 965,314 daily Agmarknet records, 2014–24 (Mendeley, CC BY 4.0). Found in our planned week-1 check.

THE REBUILD

96 markets

Maharashtra arrival-weighted monthly modal price: 85,346 clean market-days → 132 months, none missing.

THE PROOF

0.99 r on levels vs NHRDF Lasalgaon, 66 months

0.93 r on monthly % changes

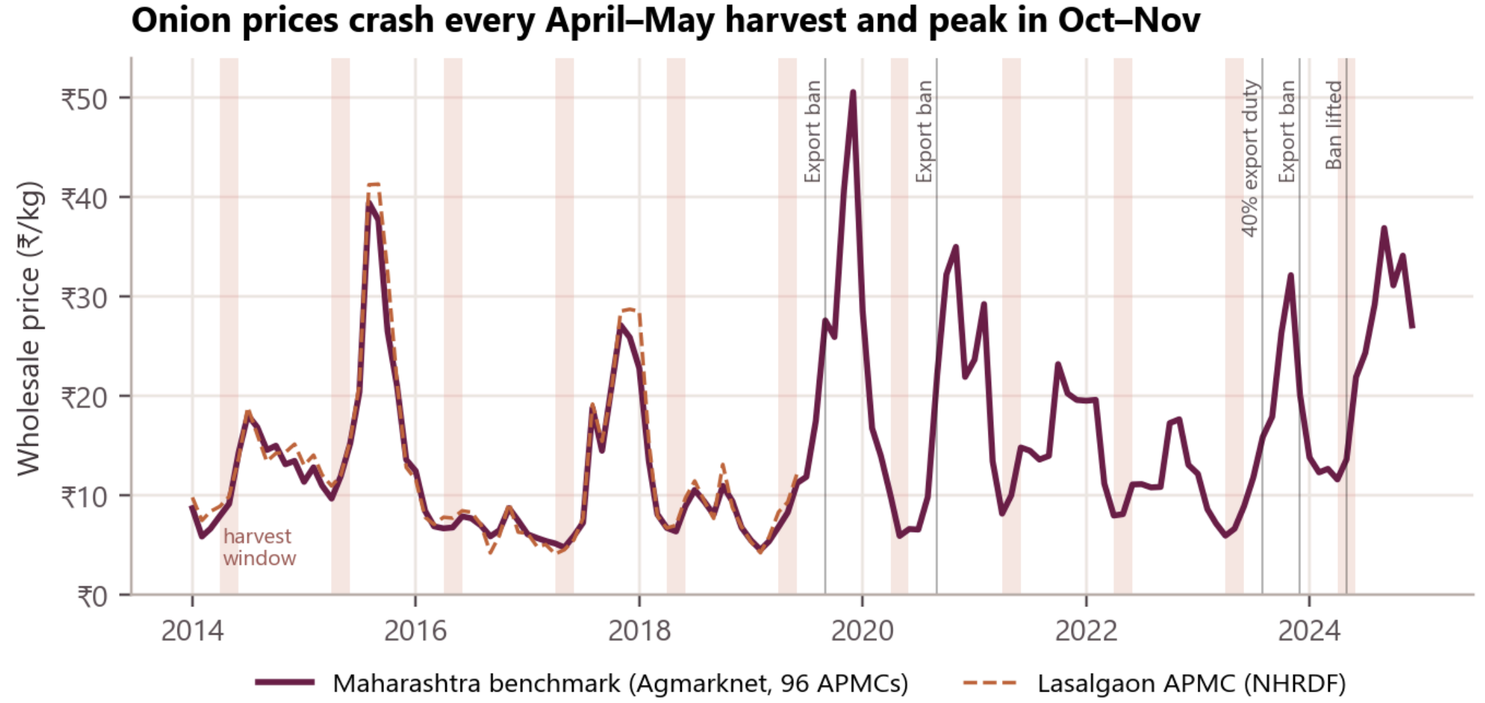
0.90 r on 19 reported prices, 2020–24; ours ≈10% lower (premium market)

Double-count fix. On all 5,672 market-days with several variety rows, each row repeated the market's total arrivals. Summing would add 1,979,857 t, so we keep one row per market-day.

Weather layer. NASA POWER monthly rain, temperature and humidity at Lasalgaon (20.15°N 74.23°E), 1996–2024: satellite-derived and reanalysis estimates, free, no key.

EXPLORATORY ANALYSIS

Prices swing four-fold within a year



AVERAGE, 2014–24

Rs 14.88/kg

coefficient of variation 60.9%

RANGE, BOTH IN 2019

Rs 4.51 → 50.56

per kg, February to December

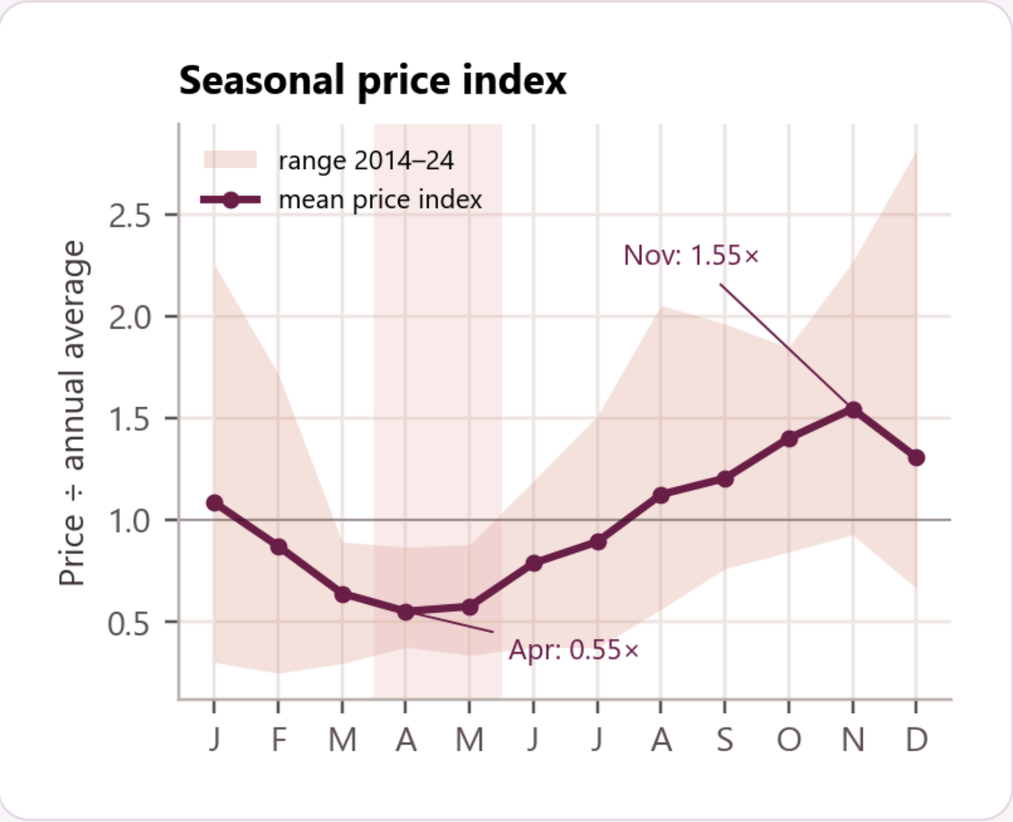
WITHIN A YEAR

4.58×

dearest month vs cheapest, on average

SEASONALITY

Crash in April, peak in November



APRIL

0.55×

NOVEMBER

1.55×

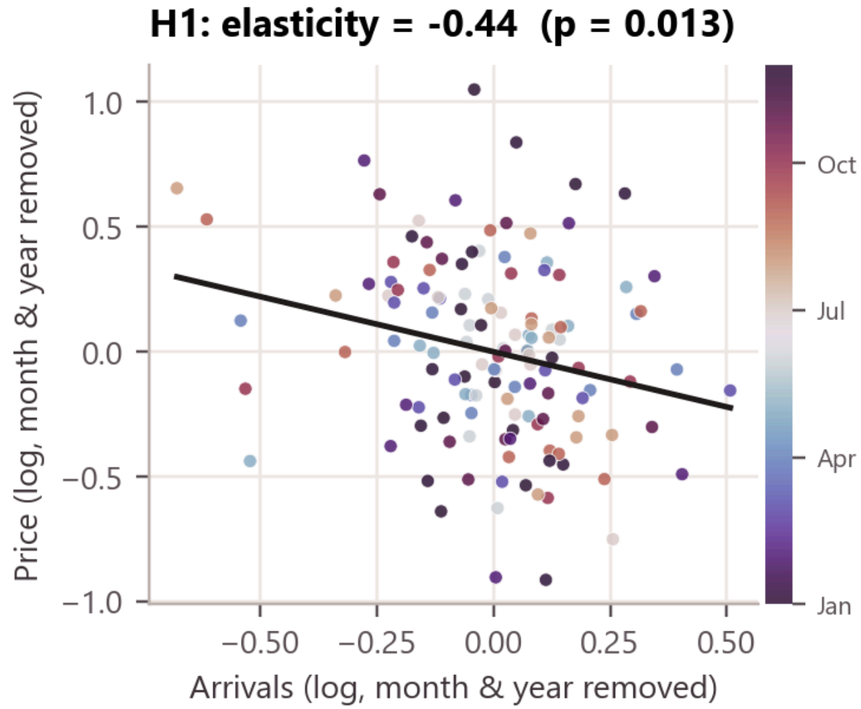
Price ÷ that year's average price, mean of 2014–24.

11 / 11 years in which November beat April

148.3% median rise from April to November

0.61× → 1.42× Lasalgaon 1996–2013; its peak is 1.49× in Oct

Heavier arrivals go with lower prices



β = −0.44

p = 0.013 (OLS = HAC)
n = 132 months
adj R^2 = 0.53

$\ln P = a + \beta \cdot \ln A$ + 11 month + 10 year dummies
A = arrivals per market-day

A month with 10% heavier arrivals has prices about 4.4% lower.

H1 SUPPORTED · ONE-SIDED P = 0.0066

Association, not causation

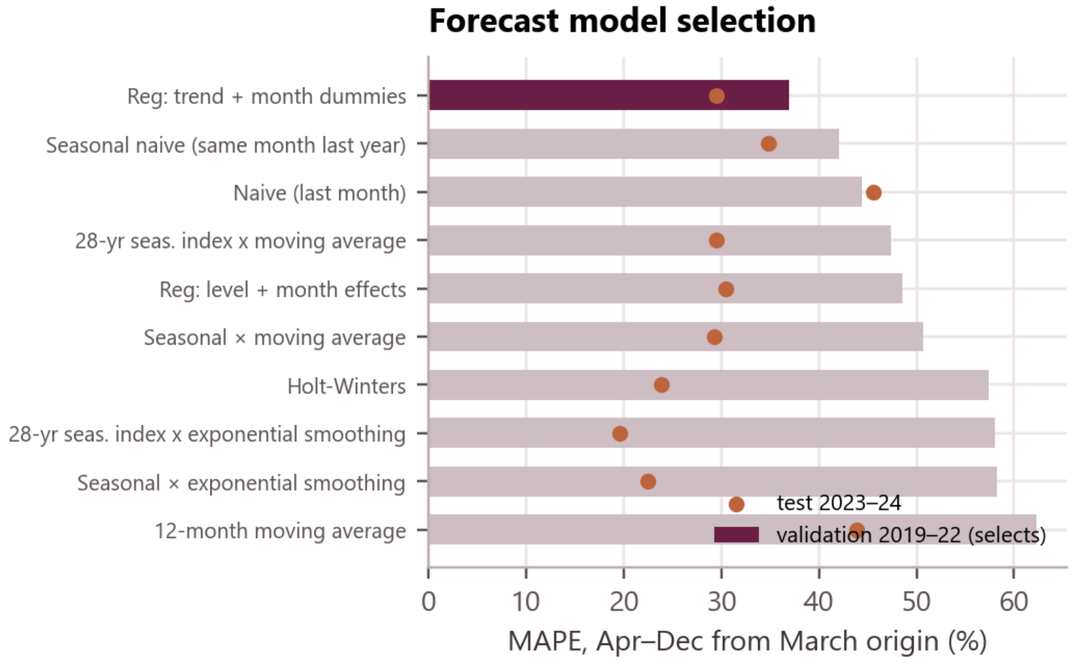
Other arrival measures: β −0.27 and −0.25 (p 0.06–0.08)

Add last month's price: effect vanishes (β −0.01, p 0.92)

Last month's price predicts arrivals (p 0.011): jointly set

FORECASTING · MODEL SELECTION

Choose on validation, score on test



1. Origin: end of March. Forecast April–December, 9 months, expanding window.
2. 10 syllabus models: naive, moving averages, exponential smoothing, Holt-Winters, regression.
3. Select the lowest Apr–Dec MAPE on validation seasons 2019–22.
4. Score once on test seasons 2023–24. No switching.

SELECTED

Regression: trend + month dummies

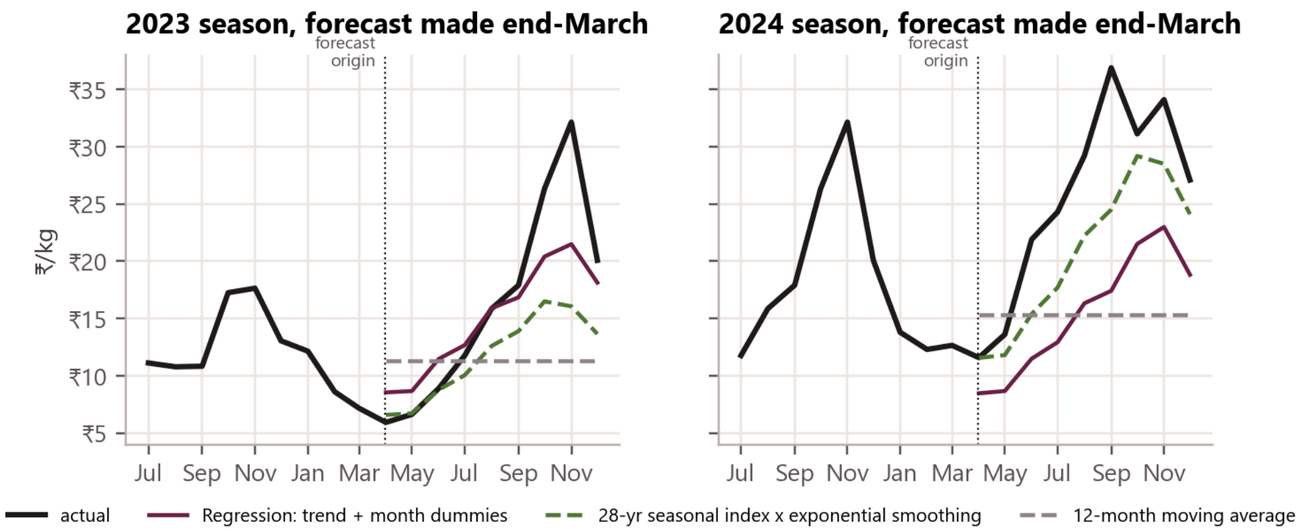
`ln P = a + b·t + 11 month dummies`

`Validation MAPE 36.94% · test 29.5%`

Best on test, not chosen: 28-yr index × smoothing, 19.55%

FORECASTING · TEST SEASONS

Right shape, level biased low



For selling, **when** prices peak matters more than **how high**. The plan's own value estimate was below what it earned in 6 of 8 seasons.

TEST MAPE, 2023–24

29.5%

Apr–Dec error, from a March origin

BIAS

−17.1%

forecasts run below actual prices

DIRECTION OF MOVES

15 / 16

monthly ups and downs called right (93.75%)

The sell-or-store linear programme

MAXIMISE

$$\Sigma\ p(t)\cdot x(t) - h\cdot\Sigma\ s(t)$$

forecast revenue minus storage cost

DECIDE, T = APR … DEC

x(t) tonnes sold
s(t) tonnes in store

SUBJECT TO

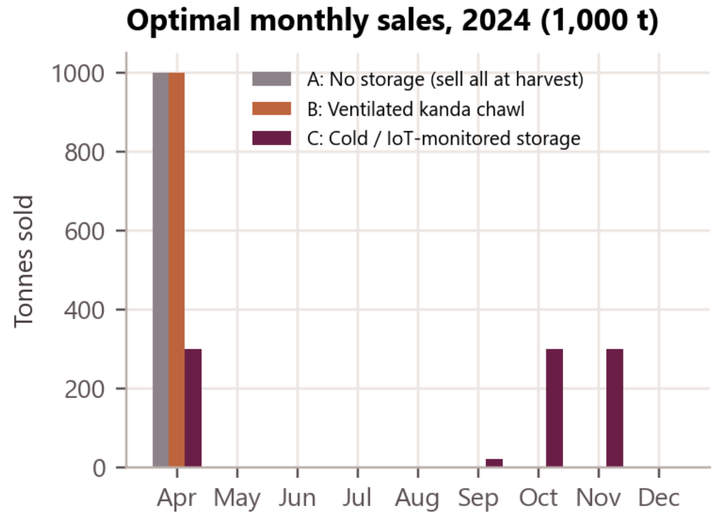
$s(1) = Q - x(1)$	April: unsold crop is stored (Q = 1,000 t)
$s(t) = (1 - \delta)\cdot s(t-1) - x(t)$	stock balance: rot δ, then sales
$s(t) \le C = 700$ t	storage capacity
$x(1) \ge L = 250$ t	crop-loan sale at harvest
$x(t) \le U = 300$ t, May–Dec	grading and trucking cap
$s(Dec) = 0;\ x,\ s \ge 0$	store empty by December

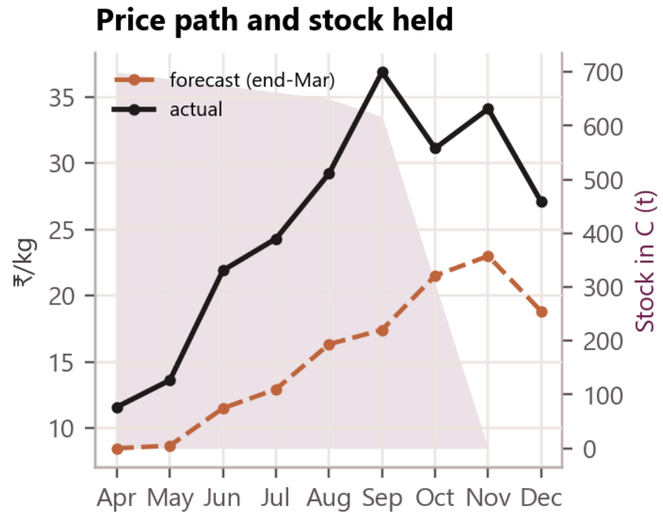
	STORAGE	LOSS δ/MO	h RS/T/MO	CAPACITY	2024 PLAN	VS A
A	None: sell all at harvest	—	—	0 t	Rs 84.7 lakh	—
B	Ventilated kanda chawl	18.05%	250	700 t	Rs 84.7 lakh	+0
C	Cold / IoT-monitored store	1.83%	700	700 t	Rs 132.5 lakh	+47.8 lakh

δ = 1 − (1 − L120)^(1/4), from ICAR 120-day losses of 54.89% and 7.14%. h is the operating cost of an owned store; capital cost is counted separately. Plan value at the end-March forecast.

OPTIMISATION · 2024 PLAN

The 2024 plan and its shadow prices





C: 300 t in April · 21 t Sep · 300 t Oct · 300 t Nov · 78.8 t lost to rot

A and B: all 1,000 t in April. At 18% a month, the forecast rise never beats the rot.

SHADOW PRICES, VERIFIED ±1 T

STORAGE CAPACITY C

+Rs 4,028

per extra tonne of space

HANDLING CAP U

+Rs 6,526

per extra t/month; binds in Oct and Nov

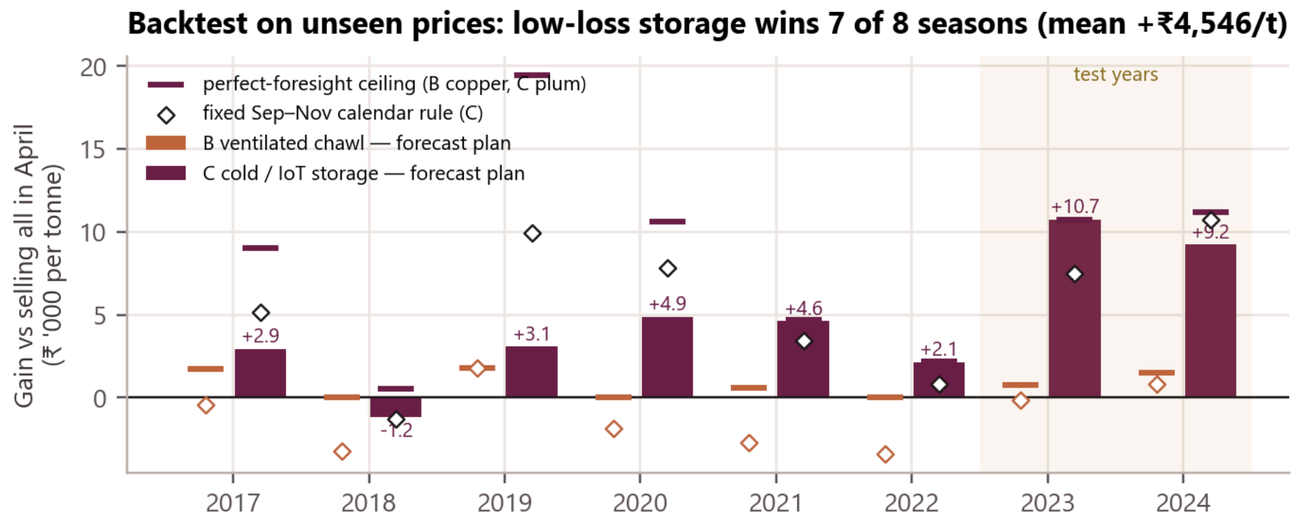
LOAN SALE L

Rs 0

slack: a 700 t store already forces 300 t out

BACKTEST · H2

Tested on prices the model never saw



Insight: a fixed Sep–Nov rule earns +Rs 5,508/t in cold storage, above the plan's +Rs 4,546/t. The seasonal shape does the work; the LP adds knowing when **not** to store.

H2 CHAWL · NOT SUPPORTED

0 of 8

seasons in which the chawl plan stored at all. A naive Sep–Nov rule in a chawl loses Rs 1,149/t.

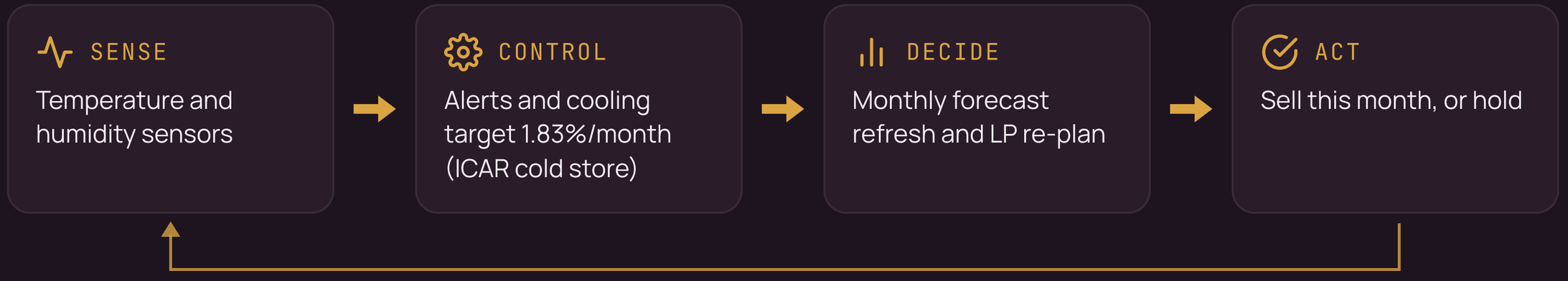
COLD / IOT · WINS 7 OF 8

+Rs 4,546/t

mean gain over harvest sale (median 3,840); paired t one-sided p = 0.006, Wilcoxon 0.008. Test seasons 2023–24: +Rs 9,980/t.

INDUSTRY 4.0 VALUE · H3 SUPPORTED

Smart storage pays its way



Rs 4,546/t

over the chawl per tonne harvested, mean of 8 seasons (Rs 4.55/kg); wins 7 of 8, paired t p = 0.006

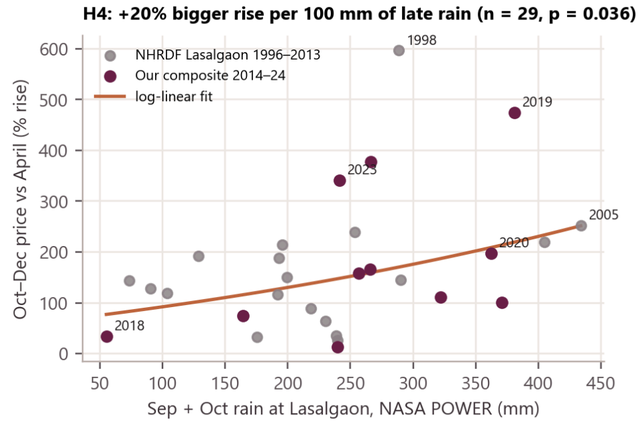
1.5 seasons

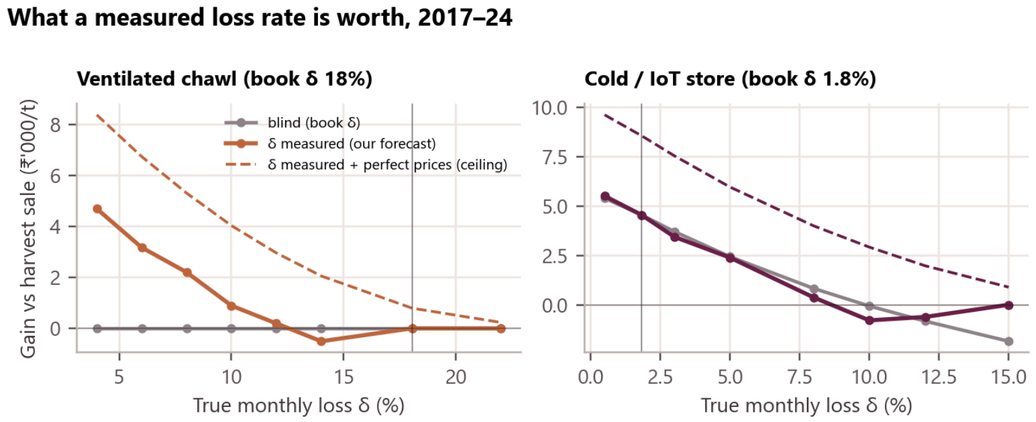
payback on the extra capital: Rs 16,667/t of space (ICAR multi-module) vs Rs 7,000/t for a chawl

Rs 45.5 lakh

a year for a 1,000 t FPO; the 2023–24 test seasons alone gave Rs 9,980/t

What satellites and sensors are worth





H4 · NASA SEP–OCT RAIN

+20%

bigger rise per 100 mm, p = 0.036: **weakly supported, fragile**

LIVE RAIN FEED

Rs 0/t

the 300 t handling cap binds: the November sale is already fixed

CHAWL MEASURED AT 8%

Rs 2,200/t

the book 18% says never store; measuring first turns storage on

COLD-STORE SENSING

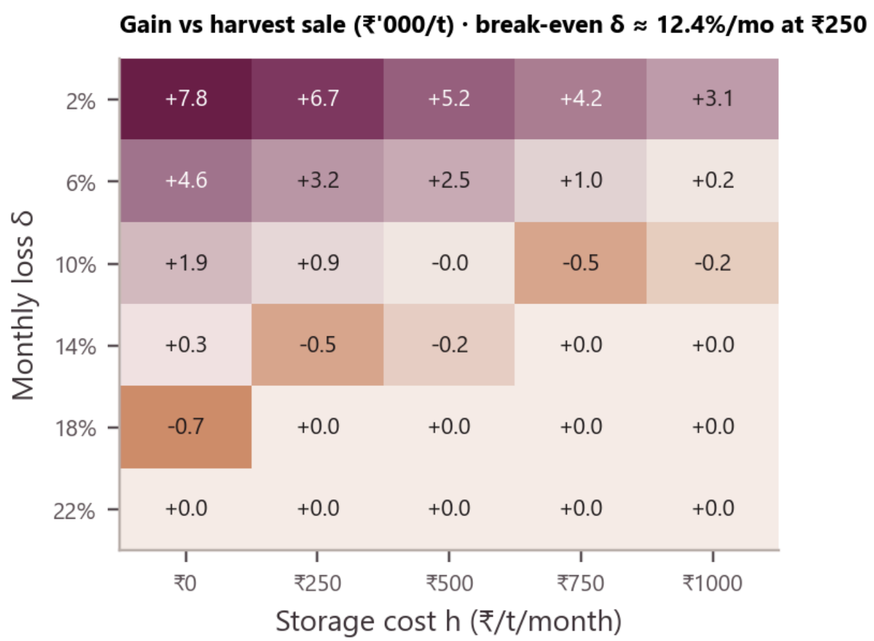
Rs 545–746/t

if loss climbs to 8–15% a month, and only with accurate prices

> **Lesson:** measure the store before the season; sensors pay only alongside a good price forecast.

SENSITIVITY

Storage pays below 12.4% loss a month



12.4% break-even monthly loss at Rs 250/t/month.
ICAR chawl: 18%. Cold store: 1.83%.

STRESS TEST, COLD STORE	MEAN GAIN
Base case: δ 1.83%, h Rs 700	+Rs 4,546/t
Loss 8% a month	+Rs 374/t
Cost doubled to Rs 1,400/t/month	+Rs 1,069/t
No handling cap (U = 1,000 t)	+Rs 4,131/t
Capacity 300 t instead of 700 t	+Rs 1,770/t

Each point of monthly loss avoided ≈ **Rs 685/t** (average, 0.5% → 8%)

MANAGERIAL IMPLICATIONS

What each decision-maker should do

FPOs

- Do not store in a chawl on hope: at 18% a month it loses money.
- With low-loss storage, sell the minimum in April, hold for Sep–Nov, re-plan monthly.
- Add capacity and handling first: +Rs 4,028 and +Rs 6,526 per tonne (2024 shadow prices).

+Rs 4,546/t

mean gain of the cold-store plan

Buffer agencies

- Storage type decides whether a buffer earns or rots: NAFED lost ≈53% in 2019.
- Buy near the April trough, release in Sep–Nov, from low-loss stores only.

0.55× → 1.55×

April vs November seasonal index

Investors and policy

- Payback ≈1.5 seasons at Rs 16,667/t; 0.15 at the MIDH Rs 8,000/t norm.
- Not viable for small cold rooms at Rs 50,000/t (−Rs 503/t a year).
- Fund loss reduction: each point of monthly loss ≈ Rs 685/t.

1.5 seasons

payback on the extra capital

LIMITATIONS

What our model does not claim

Composite, not Lasalgaon

Maharashtra price runs ≈10% below Lasalgaon, so Lasalgaon gains would be higher: Nasik gives +Rs 11,510 and +Rs 13,377/t in 2023–24.

H1 and H4 are associations

H1 weakens with other arrival measures (p 0.06–0.08) and vanishes with last month's price. H4 (rain) is fragile: drop 2005, 2018 or 2019 and p > 0.05.

Forecasts biased low

Test MAPE 29.5%, bias −17%. Value comes from the seasonal shape; a fixed Sep–Nov rule does as well or better (+Rs 5,508/t).

Constant loss rates

ICAR 120-day trials extrapolated to 8 months; monsoon sprouting may raise losses. At 8% a month the cold store still gains.

Not modelled

Quality grading, credit beyond L, export bans (2019, 2020, 2023). Price impact is small: the FPO is 1.8–3.6% of arrivals.

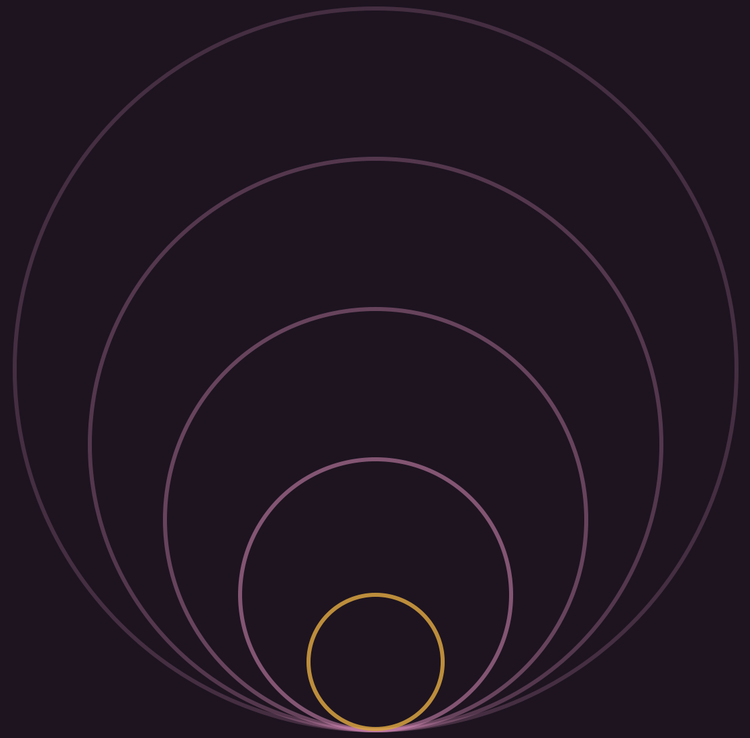
Small sample

8 seasons, 2 of them test seasons: the tests are indicative. 2018 was a losing year for cold storage (−Rs 1,181/t).

THE ANSWER

Cut the rot first, then time the sale.

Low-loss storage adds Rs 4.55 a kilo over a chawl on average, 2017–24, and pays back in about 1.5 seasons. Measure the store's real loss rate before the season.



Questions कांदा

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